

An Interactive Framework for Named Entity-Based Reproducibility Analysis of Bioinformatics Workflows

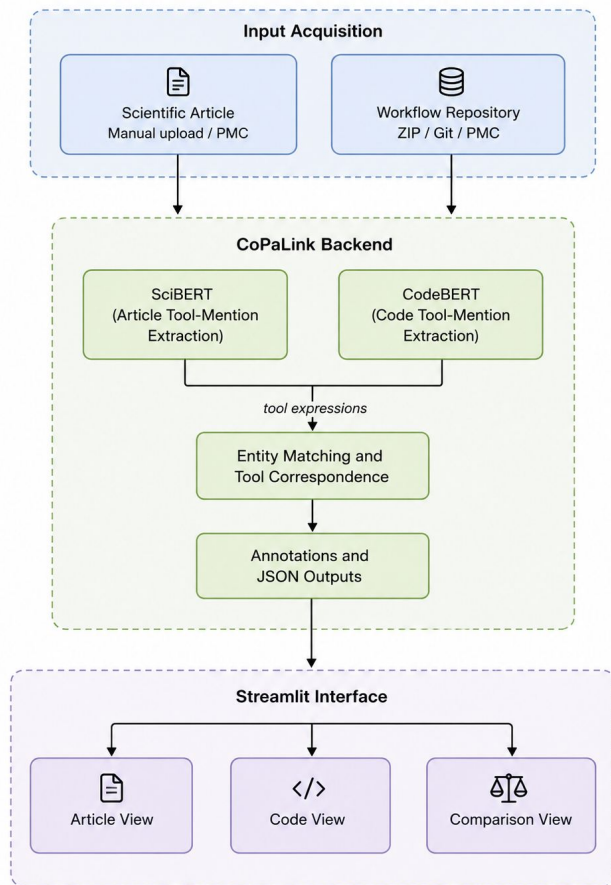
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Bioinformatics workflows combine diverse tools with inconsistent naming across sources



Publication and workflow code
comparison after tool extraction using bert model.

Interactive Streamlit interface for output
visualization

Similarity and coverage statistics are also calculated and shown in
the interface.

Workflow comparison through an interactive interface

Choose view

☒ Comparison view ☐ Article view ☐ Code view

Select tool to compare

All common tools

Article View

in FASTQ or FASTA format (potentially raw, untrimmed reads, see Section 2.3), corresponding to the metagenomic or transcriptomic dataset(s) to query [Fig. 1(1)].

** One or several homologous groups of reference proteins, each sequence annotated with its taxonomic affiliation (in the EggNOG format, containing NCBI's taxonomy ID and a unique identifier, e.g. '70448.Q0P3H7').

2.1 Alignment and tree reconstruction of references

For each input group of reference sequences a multiple sequence alignment is computed using either MAFFT (Kato and Standley, 2013) or PRANK (Löytynoja and Goldman, 2010), without applying any filtering or trimming methods. Then a reference tree is reconstructed using any of IQ-TREE (Nguyen et al., 2015), RAXML-NG (Kozlov et al., 2019; Stamatakis, 2014) or FastTree (Price et al., 2010), making it possible to choose the appropriate method for a specific analysis. This way the

Code View ↔

Repository file

main.nf — alignReferences

```
output:
  file "${fasta.baseName}.aln" into ref_alignments
  file "${fasta.baseName}.alignment.log" into
alignment_logs

publishDir "${params.reference_dir}/
${fasta.simpleName}", mode: 'copy'
tag "${fasta.simpleName}"
stageInMode 'copy'

script:
  if (params.align_method == "prank")
    ""
    prank -protein -d=$fasta -o=${fasta.baseName} -
f=fasta
    sed -i '/^>!/ s/[U|*|X]/-/g'
${fasta.baseName}.best.fas
```

Thank you!

Learn more about the workflow comparison and its results on my poster.



Please scan for more details

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